

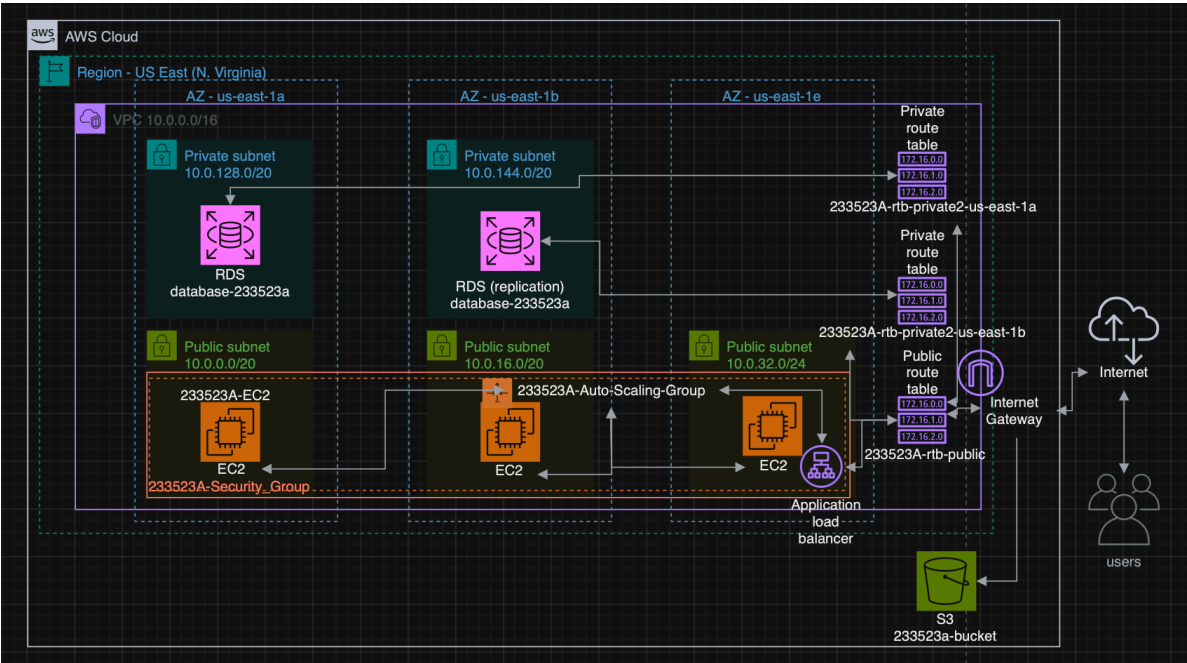


2024/2025 SEMESTER 1 – PROJECT

Course : Diploma in AI and Data Engineering

Module : EGT208 - Cloud Computing & Platform

1. Cloud Architecture



2. Explain the functions of the web application in detail.

Company XYZ sells electronic appliances such as fridges, TVs, and washing machines. They are currently operating on an on-premise server, but it is not going too well. The company is having trouble with the monthly payments for professionals to maintain the servers and keep the infrastructure up-to-date. It is also difficult for them to predict the ever-changing demand and be prepared for it.

Thus, the company has decided to migrate its server to the AWS cloud and entrusted the team at DADE AI with the process.

Below is the proposal by DADE AI to carry out the process:

On-Premise	AWS Cloud
------------	-----------

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The web application is written in python using Python Flask to render the html templates. Gunicorn is then used to facilitate Python Flask functions by efficiently managing multiple simultaneous requests. The workload among Gunicorn workers is balanced and distributed equally by Nginx.	The same program can be run on an AWS EC2 instance by setting up the environment and installing necessary packages.
A number of local computers are currently used to host the web server.	EC2 or Elastic Compute Cloud is a cloud-based computer that can be configured and scaled in or out within minutes. It works in the same way as a local computer. Currently, one EC2 instance is created in the us-east-1a Availability Zone.
The company operates globally, and it is essential that web server is accessible by customers at low latency. So, the company has set up their own networks in different places.	AWS Virtual Private Clouds are separated from the AWS cloud and can be configured as needed. A VPC can span over multiple AZ , allowing the web server to be accessible from all across the world.
A Firewall has been set up to protect and monitor the network.	A VPC is attached to a Security Group , which is a Firewall equivalent in AWS. 233523A-Security_Group has three inbound rules: HTTP, HTTPS and MySQL, meaning only these three requests are permitted and the rest are denied thereby protecting the network.
As the web application needs to display item information and images to customers online, the company also has physical data centres to store the data although it is on a small scale.	Relational Database or RDS by AWS functions in the same way as a local hosted RDBMS; it stores tables in databases, i.e., information about products. An EC2 instance can retrieve the data through MySQL queries in this case. S3 bucket is a Simple Storage Service by AWS that can store data in any format. The data is accessible via links, and currently images for the products are securely stored in 233523A-bucket.

On top of the equivalent features, Company XYZ can benefit greatly by migrating to the AWS Cloud, as the AWS Cloud Infrastructure is based on the AWS well-architected framework.

Operational Excellence

- Perform operation as code: AWS CLI can be used to do anything that is done on the AWS console. Managing AWS through codes automate and streamline the process while reducing human error.

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Admin No. 233523A

I have tried but this cannot be tested as a new IAM role for my computer cannot be created. Here is an example of how the same EC2 can be created through code.

```

pip install awscli --upgrade --user
aws --version
aws ec2 create-key-pair --key-name 233523A-Key_Pair --query 'KeyMaterial'
--output text > 233523A-Key_Pair.pem
chmod 400 233523A-Key_Pair.pem
aws ec2 run-instances \
--image-id ami-0de53d8956e8dcf80 \ # This is a common Ubuntu AMI in us-east-1, replace if needed.
--instance-type t2.micro \
--key-name 233523A-Key_Pair \
--security-group-ids YOUR_GROUP_ID \
--subnet-id YOUR_PUBLIC_SUBNET_ID \
--tag-specifications
'ResourceType=instance,Tags=[{Key=Name,Value=233523A-EC2}]'

```

Security

- Apply security at all layers
AWS allows security policies to be attached to individuals and services. In this case, the EC2 retrieving images from S3 can be given a read-only policy and performing CRUD operations RDS-full-access instead of giving full access to all the resources.

Reliability

- RDS Multi-AZ deployment
Deploying RDS in multiple AZs ensures higher reliability, even during system updates and downtime of one zone.
- Elastic Load Balancing
ELB distributes the incoming load equally among the existing EC2 instances to ensure a smooth customer experience.
- Auto Scaling Groups
ASG facilitates ELB by regularly checking the health of EC2 instances, deleting and creating new instances as indicated or on-demand traffic.

Performance Efficiency

- Elastic Beanstalk
Elastic Beanstalk is a managed service that combines EC2, ELB and ASG. Implementing this will be a lot simpler and more efficient than individually creating the three services. But, I have tried this, and it is not allowed.
- AWS Lambda
AWS Lambda is useful for serverless computing where predefined functions are triggered by certain events. In the web application, instead of hosting an EC2 to listen for incoming requests and perform CRUD operations, writing lambda functions for those operations will be more efficient as Lambda offers free computing up to a certain extent. To do so, the function itself must be given an IAM role to access the required resources, which is not allowed. Here is how 'select' would look like:

```

• import pymysql
•

```

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Admin No. 233523A

```

• # Retrieve database connection details from environment variables
• endpoint = "database-233523a.croh44znaqo.us-east-1.rds.amazonaws.com"
• username = "admin"
• password = "Qwas1357"
• database_name = "db1"
•
•
• try:
•     # Connect to the database
•     conn = pymysql.connect(host=endpoint, user=username, password=password,
•                             database=database_name)
•
• except pymysql.MySQLError as e:
•     print('statusCode: 500')
•     print(f"Error connecting: {str(e)}")
•     conn = False
•
•
• def handler():
•     if conn:
•         if conn.open:
•             cursor = conn.cursor()
•             query = "SELECT * FROM products"
•             cursor.execute(query)
•             conn.commit()
•             rows = cursor.fetchall()
•
•             for row in rows:
•                 print("{0} {1} {2} {3}".format(row[0], row[1], row[2],
• row[3]))
•         else:
•             return
•
• handler()

```

- AWS CloudFront
AWS CloudFront is a Content Delivery Network that delivers static and dynamic web content to users. It is similar to S3 but CloudFront offers a certain amount of free resources which can be more cost-effective and reliable for web pages.

Cost Optimization

- AWS CloudWatch
AWS CloudWatch has alarms that send notifications to registered emails upon some events. In this case, if \$40 out of the budget \$50 is spent, an email will be immediately sent to 233523A@mymail.nyp.edu.sg to take necessary actions.

Sustainability

- Stop RDS and EC2 when not in use
This is more of an ethical and responsible use of the resources and not abuse them.

Appendix A

233523A-vpc

aws

Services

Search

[Option+S]

N. Virginia

voclabs/user3238427=MIN_PHYO_THURA @ 2517-7295-6993

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

Endpoints

Endpoint services

NAT gateways

Peering connections

VPC > Your VPCs > vpc-0fcb8ea5e50847cb4

vpc-0fcb8ea5e50847cb4 / 233523A-vpc

Actions

Details

Info

VPC ID

vpc-0fcb8ea5e50847cb4

Tenancy

Default

Default VPC

No

Network Address Usage metrics

Disabled

State

Available

DHCP option set

dopt-033729f26c719bfa8

IPv4 CIDR

10.0.0.0/16

Route 53 Resolver DNS Firewall rule groups

Failed to load rule groups

DNS hostnames

Enabled

Main route table

rtb-0c4c79f2a7b0b7109

IPv6 pool

-

Owner ID

251772956993

DNS resolution

Enabled

Main network ACL

acl-09c3875f8027fb6db

IPv6 CIDR (Network border group)

-

Resource map

CIDRs

Flow logs

Tags

Integrations

Resource map

Info

VPC

Show details

Your AWS virtual network

Subnets (4)

Subnets within this VPC

us-east-1a

233523A-subnet-public1-us-east-1a

233523A-subnet-private1-us-east...

us-east-1b

233523A-subnet-public2-us-east-1b

233523A-subnet-private2-us-east...

Route tables (4)

Route network traffic to resources

233523A-rtb-private1-us-east-1a

233523A-rtb-private2-us-east-1b

rtb-0c4c79f2a7b0b7109

233523A-rtb-public

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Admin No. 233523A

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Services

Search

[Option+S]

N. Virginia

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VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

Endpoints

Endpoint services

NAT gateways

Peering connections

VPC > Security Groups > sg-0e84a4f8e47a92046 - 233523A-Security_Group

sg-0e84a4f8e47a92046 - 233523A-Security_Group

Actions

Details

Security group name

233523A-Security_Group

Security group ID

sg-0e84a4f8e47a92046

Description

Security Group

VPC ID

vpc-0fcb8ea5e50847cb4

Owner

251772956993

Inbound rules count

2 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Tags

Inbound rules (2)

Manage tags

Edit inbound rules

Search

< 1 >

	Name	Security group rule...	IP version	Type	Protocol
☐					

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Services

Search

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VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

Endpoints

Endpoint services

NAT gateways

Peering connections

VPC > Security Groups > sg-0e84a4f8e47a92046 - 233523A-Security_Group

sg-0e84a4f8e47a92046 - 233523A-Security_Group

Actions

Details

Security group name

233523A-Security_Group

Security group ID

sg-0e84a4f8e47a92046

Description

Security Group

VPC ID

vpc-0fcb8ea5e50847cb4

Owner

251772956993

Inbound rules count

2 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Tags

Inbound rules (2)

Manage tags

Edit inbound rules

Search

< 1 >

	Name	Security group rule...	IP version	Type	Protocol
☐	-	sgr-0e490fc9fedb57be6	IPv4	SSH	TCP
☐	-	sgr-0dbc134fa2da1d4aa	IPv4	HTTP	TCP

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Admin No. 233523A

aws

Services

Search

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EC2 Dashboard

EC2 Global View

Events

Console-to-Code

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

EC2 > Instances > i-04a57fd9687bd0832

Instance summary for i-04a57fd9687bd0832 (233523A-EC2)

Connect

Instance state

Actions

Updated less than a minute ago

Instance ID

i-04a57fd9687bd0832 (233523A-EC2)

Public IPv4 address

52.90.120.255 | open address

Private IPv4 addresses

10.0.14.208

IPv6 address

-

Instance state

Running

Public IPv4 DNS

ec2-52-90-120-255.compute-1.amazonaws.com | open address

Hostname type

IP name: ip-10-0-14-208.ec2.internal

Private IP DNS name (IPv4 only)

ip-10-0-14-208.ec2.internal

Elastic IP addresses

-

Answer private resource DNS name

-

Instance type

t2.micro

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto-assigned IP address

52.90.120.255 [Public IP]

VPC ID

vpc-0fcb8ea5e50847cb4 (233523A-vpc) | open address

Auto Scaling Group name

-

IAM Role

-

Subnet ID

subnet-026158b18b453a4c8 (233523A-subnet) | open address

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Services

Search

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EC2 Dashboard

EC2 Global View

Events

Console-to-Code

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Security details

IAM Role

-

Owner ID

251772956993

Launch time

Tue Jul 30 2024 14:56:12 GMT+0800 (Singapore Standard Time)

Security groups

sg-0e84a4f8e47a92046 (233523A-Security_Group) | open address

Inbound rules

Filter rules

< 1 >

Name	Security group rule ID	Port range	Protocol	Source
-	sgr-0e490fc9fedb57be6	22	TCP	0.0.0.0/0
-	sgr-0dbc134fa2da1d4aa	80	TCP	0.0.0.0/0

Outbound rules

Filter rules

< 1 >

Name	Security group rule ID	Port range	Protocol	Destination
------	------------------------	------------	----------	-------------

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Services

Search

[Option+S]

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EC2 Dashboard

EC2 Global View

Events

Console-to-Code

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

CloudShell

Feedback

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Networking details info

Public IPv4 address
52.90.120.255 | open address

Private IPv4 addresses
10.0.14.208

VPC ID
vpc-0fcb8ea5e50847cb4 (233523A-vpc)

Public IPv4 DNS
ec2-52-90-120-255.compute-1.amazonaws.com | open address

Private IP DNS name (IPv4 only)
ip-10-0-14-208.ec2.internal

Subnet ID
subnet-026158b18b453a4c8 (233523A-subnet-public1-us-east-1a)

IPv6 addresses
-

Secondary private IPv4 addresses
-

Availability zone
us-east-1a

Carrier IP addresses (ephemeral)
-

Outpost ID
-

Use RBN as guest OS hostname
Disabled

Answer RBN DNS hostname IPv4
Disabled

▼ Network Interfaces (1) Info

Filter network interfaces

CloudShell

Feedback

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233523A-Security_Group

aws

Services

Search

[Option+S]

N. Virginia

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VPC dashboard

EC2 Global View

Filter by VPC

▼ Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

Endpoints

Endpoint services

NAT gateways

Peering connections

CloudShell

Feedback

Details

Security group name
233523A-Security_Group

Security group ID
sg-0e84a4f8e47a92046

Description
Security Group

VPC ID
vpc-0fcb8ea5e50847cb4

Owner
251772956993

Inbound rules count
3 Permission entries

Outbound rules count
1 Permission entry

Inbound rules

Outbound rules

Tags

Inbound rules (3)

Search

Manage tags

Edit inbound rules

1

	Name	Security group rule...	IP version	Type	Protocol
☐	-	sgr-0e490fc9fedb57be6	IPv4	SSH	TCP
☐	-	sgr-0dbc134fa2da1d4aa	IPv4	HTTP	TCP
☐	-	sgr-06e3538272bf5caa3	IPv4	MYSQL/Aurora	TCP

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233523A-subnet

aws

Services

Search

[Option+S]

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Amazon RDS

Dashboard

Databases

Query Editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

Events

Event subscriptions

RDS > Subnet groups > subnet-233523a

subnet-233523a

Subnet group details

VPC ID

vpc-0fcb8ea5e50847cb4

ARN

arn:aws:rds:us-east-1:251772956993:subgrp:subnet-233523a

Supported network types

IPv4

Description

subnet

Subnets (4)

Availability zone	Subnet ID	CIDR block
us-east-1b	subnet-062c2dbb3e4420833	10.0.144.0/20
us-east-1a	subnet-026158b18b453a4c8	10.0.0.0/20

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Services

Search

[Option+S]

N. Virginia

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Amazon RDS

Dashboard

Databases

Query Editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

Events

Event subscriptions

Description

subnet

Subnets (4)

Availability zone	Subnet ID	CIDR block
us-east-1b	subnet-062c2dbb3e4420833	10.0.144.0/20
us-east-1a	subnet-026158b18b453a4c8	10.0.0.0/20
us-east-1b	subnet-081623510f4e943a1	10.0.16.0/20
us-east-1a	subnet-0e4894b8fffcf248f	10.0.128.0/20

Tags (0)

Manage tags

Filter by Tag key

< 1 >

Tags

Value

You don't have any tags associated with this resource.

Manage tags

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Admin No. 233523A

database-233523A

aws

Services

Search

[Option+S]

N. Virginia

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Amazon RDS

Dashboard

Databases

Query Editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

Events

Event subscriptions

RDS > Databases > database-233523a

database-233523a

Summary

DB identifier database-233523a	Status Available	Role Instance	Engine MySQL Community	Recommendations 2 Informational
CPU 3.55%	Class db.t3.micro	Current activity 0	Region & AZ us-east-1b	

Connectivity & securityMonitoringLogs & eventsConfigurationZero-ETL integrationsMail

Connectivity & security

Endpoint & port	Networking	Security
Endpoint database-233523a.crob h44znaqo.us-east-1.rds.ama	Availability Zone us-east-1b	VPC security groups 233523A-Security_Group (sg-0e84a4f8e47a92046)

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Services

Search

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Amazon RDS

Dashboard

Databases

Query Editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

Events

Event subscriptions

Connectivity & security

Endpoint & port

Networking

Security

Endpoint database-233523a.crob h44znaqo.us-east-1.rds.ama zonaws.com	Availability Zone us-east-1b	VPC security groups 233523A-Security_Group (sg-0e84a4f8e47a92046)
Port 3306	VPC 233523A-vpc (vpc-0fcb8ea5e50847cb4)	Publicly accessible Yes
	Subnet group subnet-233523a	Certificate authority rds-ca-rsa2048-g1
	Subnets subnet-062c2dbb3e4420833 subnet-026158b18b453a4c8 subnet-081623510f4e943a1 subnet-0e4894b8ffcf248f	Certificate authority date May 26, 2061, 07:34 (UTC+08:00)
	Network type IPv4	DB instance certificate expiration date July 16, 2025, 15:45 (UTC+08:00)

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Admin No. 233523A

aws

Services

Search

[Option+S]

N. Virginia

voclabs/user3238427=MIN_PHYO_THURA @ 2517-7295-6993

Amazon RDS

Dashboard

Databases

Query Editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

Events

Event subscriptions

Instance

Configuration

DB instance ID

database-233523a

Engine version

8.0.35

RDS Extended Support

Disabled

DB name

db1

License model

General Public License

Option groups

default:mysql-8-0

In sync

Amazon Resource Name (ARN)

arn:aws:rds:us-east-1:251772956993:db:database-

Instance class

Instance class

db.t3.micro

vCPU

2

RAM

1 GB

Availability

Master username

admin

Master password

IAM DB authentication

Not enabled

Multi-AZ

No

Storage

Encryption

Enabled

AWS KMS key

aws/rds

Storage type

General Purpose SSD (gp2)

Storage

20 GiB

Provisioned IOPS

-

Storage throughput

-

Storage autoscaling

Enabled

Maximum storage threshold

1000 GiB

Performance Insights

Performance Insights

enabled

Turned off

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Cookie preferences

233523A-bucket

aws

Services

Search

[Option+S]

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Amazon S3

Buckets

Access Grants

Access Points

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

IAM Access Analyzer for S3

Block Public Access settings for this account

Storage Lens

Dashboards

Storage Lens groups

AWS Organizations settings

Feature spotlight

233523a-bucket

Info

Objects

Properties

Permissions

Metrics

Management

Access Points

Objects (12)

Info

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Find objects by prefix

1

	Name	Type	Last modified	Size	Storage class
	bedroom.png	png	July 16, 2024, 15:55:43 (UTC+08:00)	2.9 MB	Standard
	coffee_machine.png	png	July 16, 2024, 15:55:41 (UTC+08:00)	264.8 KB	Standard
	kitchen.png	png	July 16, 2024, 15:55:40 (UTC+08:00)	2.3 MB	Standard

CloudShell

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index.html

aws

Services

Q Search

[Option+S]

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GNU nano 7.2

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Product Listing</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <!-- Bootstrap 4 CSS -->
  <link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css">
  <!-- Fontawesome CSS -->
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/5.15.3/css/all.min.css">
</head>
<body>
  <div class="container my-5">
    <h2 class="text-center mb-5">Product Listing</h2>
    <div class="row">
      <!-- for product in products -->
      <div class="col-lg-4 col-md-6 mb-4">
        <div class="card h-100">
          <a href="#">
            <h4 class="card-title">
              <a href="#">{{ product['product_name'] }}</a>
            </h4>
            <h5>{{ product['product_price'] }}</h5>
            <p class="card-text">{{ product['product description'] }}</p>
          </div>
        </div>
      </div>
    </div>
  </div>
</body>
</html>
```

Read 36 lines

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy

i-04a57fd9687bd0832 (233523A-EC2)

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app.py

aws

Services

Q Search

[Option+S]

N. Virginia

voclabs/user3238427=MIN_PHYO_THURA @ 2517-7295-6993

GNU nano 7.2

app.py

```
from flask import Flask, render_template
import json
import pymysql

app = Flask(__name__)

@app.route('/create')
def create():
    endpoint = "database-233523a.crohh44znaqo.us-east-1.rds.amazonaws.com"
    username = "admin"
    password = "Qwas1357"
    database_name = "db1"

    try:
        conn = pymysql.connect(host=endpoint, user=username, password=password, database=database_name)

        if conn.open:
            cursor = conn.cursor()
            query = f"CREATE TABLE products (product_id INT NOT NULL AUTO_INCREMENT, product_name VARCHAR(255) NOT NULL,
product_price DECIMAL(10,2) NOT NULL, product_description TEXT NOT NULL, PRIMARY KEY (product_id));"
            cursor.execute(query)
            conn.commit()
            return f"Table products is created."
    except pymysql.Error as e:
        return f"There is an issue in creating Table products. {e}"
```

Read 149 lines

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy

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Project / EGT208

Admin No. 233523A

awsServicesSearch[Option+S]

N. Virginiavoclabs/user3238427=MIN_PHYO_THURA @ 2517-7295-6993

GNU nano 7.2app.py

```
        conn.commit()
        return f"Table products is created."
    except pymysql.Error as e:
        return f"There is an issue in creating Table products. {e}"

@app.route('/delete')
def delete():
    endpoint = "database-233523a.crobb44znago.us-east-1.rds.amazonaws.com"
    username = "admin"
    password = "Qwas1357"
    database_name = "db1"

    try:
        conn = pymysql.connect(host=endpoint, user=username, password=password, database=database_name)

        if conn.open:
            cursor = conn.cursor()
            query = f"DROP TABLE products;"
            cursor.execute(query)
            conn.commit()
            return f"Table products is deleted."
    except pymysql.Error as e:
        return f"There is an issue in deleting Table products. {e}"

@app.route('/insert/<product_name>/<float:product_price>/<string:product_description>')
def insert(product_name, product_price, product_description):
    endpoint = "database-233523a.crobb44znago.us-east-1.rds.amazonaws.com"
    username = "admin"
    password = "Qwas1357"
    database_name = "db1"

    try:
        conn = pymysql.connect(host=endpoint, user=username, password=password, database=database_name)

        if conn.open:
            cursor = conn.cursor()
            query = f"INSERT INTO products (product_name, product_price, product_description) VALUES ({product_name}, {product_price}, {product_description})"
            cursor.execute(query)
            conn.commit()
            return f"Product {product_name} is inserted."
    except pymysql.Error as e:
        return f"There is an issue in inserting Product {product_name}. {e}"

@app.route('/update/<int:product_id>/<product_name>/<float:product_price>/<string:product_description>')
def update(product_id, product_name, product_price, product_description):
    endpoint = "database-233523a.crobb44znago.us-east-1.rds.amazonaws.com"
    username = "admin"
```

i-04a57fd9687bd0832 (233523A-EC2)

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awsServicesSearch[Option+S]

N. Virginiavoclabs/user3238427=MIN_PHYO_THURA @ 2517-7295-6993

GNU nano 7.2app.py

```
        return f"There is an issue in deleting Table products. {e}"

@app.route('/insert/<product_name>/<float:product_price>/<string:product_description>')
def insert(product_name, product_price, product_description):
    endpoint = "database-233523a.crobb44znago.us-east-1.rds.amazonaws.com"
    username = "admin"
    password = "Qwas1357"
    database_name = "db1"

    try:
        conn = pymysql.connect(host=endpoint, user=username, password=password, database=database_name)

        if conn.open:
            cursor = conn.cursor()
            query = f"INSERT INTO products (product_name, product_price, product_description) VALUES ({product_name}, {product_price}, {product_description})"
            cursor.execute(query, (product_name, product_price, product_description))
            conn.commit()
            return f"Product {product_name} is inserted."
    except pymysql.Error as e:
        return f"There is an issue in inserting Product {product_name}. {e}"

@app.route('/update/<int:product_id>/<product_name>/<float:product_price>/<string:product_description>')
def update(product_id, product_name, product_price, product_description):
    endpoint = "database-233523a.crobb44znago.us-east-1.rds.amazonaws.com"
    username = "admin"
```

i-04a57fd9687bd0832 (233523A-EC2)

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Admin No. 233523A

awsServicesSearch[Option+S]

N. Virginiavoclabs/user3238427=MIN_PHYO_THURA @ 2517-7295-6993

GNU nano 7.2app.py

```
except pymysql.Error as e:
    return f"There is an issue in inserting Product {product_name}. {e}"

@app.route('/update/<int:product_id>/<product_name>/<float:product_price>/<string:product_description>')
def update(product_id, product_name, product_price, product_description):
    endpoint = "database-233523a.crobb44znaqo.us-east-1.rds.amazonaws.com"
    username = "admin"
    password = "Qwas1357"
    database_name = "db1"

    try:
        conn = pymysql.connect(host=endpoint, user=username, password=password, database=database_name)

        if conn.open:
            cursor = conn.cursor()
            query = "UPDATE products SET product_name = %s, product_price = %s, product_description = %s WHERE product_id = %s"
            cursor.execute(query, (product_name, product_price, product_description, product_id))
            conn.commit()
            return f"Product {product_name} is updated."
    except pymysql.Error as e:
        return f"There is an issue in updating Product {product_name}. {e}"

@app.route('/select')
def select():
    endpoint = "database-233523a.crobb44znaqo.us-east-1.rds.amazonaws.com"
```

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i-04a57fd9687bd0832 (233523A-EC2)

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awsServicesSearch[Option+S]

N. Virginiavoclabs/user3238427=MIN_PHYO_THURA @ 2517-7295-6993

GNU nano 7.2app.py

```
@app.route('/select')
def select():
    endpoint = "database-233523a.crobb44znaqo.us-east-1.rds.amazonaws.com"
    username = "admin"
    password = "Qwas1357"
    database_name = "db1"
    products = []

    try:
        conn = pymysql.connect(host=endpoint, user=username, password=password, database=database_name)

        if conn.open:
            cursor = conn.cursor()
            query = "SELECT * FROM products"
            cursor.execute(query)
            conn.commit()
            rows = cursor.fetchall()

            products = []
            product_id = ''
            product_name = ''
            product_description = ''

            for row in rows:
```

HelpWrite OutWhere IsCutExecuteLocationUndoSet MarkExitRead FileReplacePasteJustifyGo To LineRedoCopy

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aws

Services

Search

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N. Virginia

voclabs/user3238427=MIN_PHYO_THURA @ 2517-7295-6993

GNU nano 7.2

app.py

```
@app.route('/')
def index():
    endpoint = "database-233523a.crobh44znaqo.us-east-1.rds.amazonaws.com"
    username = "admin"
    password = "Qwas1357"
    database_name = "db1"
    products = []

    try:
        conn = pymysql.connect(host=endpoint, user=username, password=password, database=database_name)

        if conn.open:
            cursor = conn.cursor()
            query = "SELECT * FROM products"
            cursor.execute(query)
            conn.commit()
            rows = cursor.fetchall()

            products = []
            product_id = ''
            product_name = ''
            product_description = ''

            for row in rows:
```

Help

Write Out

Where Is

Cut

Execute

Location

Undo

Set Mark

Exit

Read File

Replace

Paste

Justify

Go To Line

Redo

Copy

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Webpage (Product 1 to 3)

← → ↺ ⚠ Not Secure 54.82.196.155 ☆ 📄 🔴 ⋮

Product Listing



Vintage TV
\$299.99
Relive the golden age of television with this stylish vintage TV, combining retro charm with modern functionality. Perfect for adding a nostalgic touch to any living space.



Refrigerator
\$699.99
Keep your food fresh and organized with this state-of-the-art refrigerator, featuring advanced cooling technology and spacious compartments. Ideal for any modern kitchen.



Speakers
\$199.99
Experience high-fidelity sound with these sleek, powerful speakers, designed to deliver crystal-clear audio for all your entertainment needs. Perfect for music lovers and audiophiles alike.

54.82.196.155/#

Webpage (Product 4 to 6)

←

→

↺

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☆

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Ⓜ

⋮



Vintage Fridge

\$499.99

Add a touch of classic elegance to your kitchen with this vintage fridge, offering a blend of timeless design and reliable cooling performance. A charming centerpiece for any home.

★★★★☆



TV

\$599.99

Enjoy your favorite shows and movies in stunning clarity with this modern TV, featuring ultra-high-definition resolution and smart connectivity. A must-have for any contemporary living room.

★★★★☆



Coffee Machine

\$149.99

Start your day with the perfect brew using this innovative coffee machine, designed for convenience and precision. Ideal for coffee enthusiasts who demand quality and ease of use.

★★★★☆

54.82.196.155/#

Webpage (Product 7 to 9)

←

→

↺

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☆

🔖

Ⓜ

⋮



Bedroom

\$899.99

Transform your sleep space into a serene retreat with this stylish bedroom set, featuring modern design and high-quality materials. Perfect for creating a restful and elegant atmosphere.

★★★★☆



Kitchen

\$1200.99

Upgrade your culinary experience with this modern kitchen setup, equipped with top-of-the-line appliances and sleek cabinetry. Designed for both functionality and aesthetic appeal.

★★★★☆



Laundry

\$799.99

Simplify your laundry routine with this efficient and stylish laundry room setup, featuring advanced washing and drying technology. Perfect for busy households seeking convenience and quality.

★★★★☆

Webpage (Product 10 to 12)



Washing Machine

\$499.99

Achieve spotless laundry results with this high-performance washing machine, designed to handle all fabric types with care. Essential for any home aiming for efficient cleaning.

★★★★☆



Office

\$999.99

Create a productive and stylish workspace with this modern office setup, featuring ergonomic furniture and smart organization solutions. Ideal for professionals seeking comfort and efficiency.

★★★★☆



Vacuum Cleaner

\$249.99

Keep your home spotless with this powerful vacuum cleaner, engineered for optimal suction and easy maneuverability. Perfect for maintaining a clean and healthy living environment.

★★★★☆

AWS Trusted Advisor [not allowed]

aws

Services

Search

[Option+S]

Global

voclabs/user3238427=MIN_PHYO_THURA @ 2517-7295-6993

Trusted Advisor

Recommendations

Recommendations

Cost optimization

Performance

Security

Fault tolerance

Service limits

Operational excellence

Preferences

Manage Trusted Advisor

Notifications

Trusted Advisor Recommendations

Refresh all checks

Download all checks

Use this page to get an overview of the check results in your AWS account. Choose a check name or category to view the recommended actions or potential issues that Trusted Advisor has identified. Each check provides more information about how to address any issues. You can also download a summary of all check results. [Learn more](#)

Checks summary

0

Action recommended

Info

0

Investigation recommended

Info

0

Checks with excluded items

Info

Recommended Actions

Upgrade your AWS Support plan to get all Trusted Advisor checks

You will have access to technical support from a cloud support engineer, with phone and chat support. You can also use the AWS Support API, allow your AWS Identity and Access Management (IAM) users to create support cases for your AWS account, receive architecture support and use case

Upgrade

CloudShell

Feedback

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233523A-AMI

EC2 Dashboard

EC2 Global View

Events

Console-to-Code

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

EC2 > Launch templates > 233523A-LabConfig-AMI-Template

233523A-LabConfig-AMI-Template (lt-0b86b51f67db041dd)

ActionsDelete template

Launch template details

Launch template ID lt-0b86b51f67db041dd	Launch template name 233523A-LabConfig-AMI-Template	Default version 1	Owner arn:aws:sts::251772956993:assumed-role/voclabs/user3238427=MIN_PHYO_THURA
--	--	----------------------	--

DetailsVersionsTemplate tags

Launch template version details

ActionsDelete template version

Version 1 (Default)	Description -	Date created 2024-08-06T05:49:17.000Z	Created by arn:aws:sts::251772956993:assumed-
------------------------	------------------	--	--

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233523A-target-group

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Auto Scaling

Auto Scaling Groups

EC2 > Target groups > 233523A-target-group

233523A-target-group

Actions

Details

arn:aws:elasticloadbalancing:us-east-1:251772956993:targetgroup/233523A-target-group/4f40907590769338

Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-0fcb8ea5e50847cb4
IP address type IPv4	Load balancer 233523A-load-balancer		
4 Total targets	2 Healthy 0 Unhealthy 0 Anomalous	1 Unused	0 Initial
1 Draining			

Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.

TargetsMonitoringHealth checksAttributesTags

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233523A – Public Route Table

aws

Services

Search

[Option+S]

N. Virginia

voclabs/user3238427=MIN_PHYO_THURA @ 2517-7295-6993

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

Endpoints

Endpoint services

NAT gateways

Peering connections

VPC > Route tables > rtb-0213d94021310d70f

rtb-0213d94021310d70f / 233523A-rtb-public

Details

Info

Route table ID

rtb-0213d94021310d70f

Main

No

Explicit subnet associations

3 subnets

Edge associations

-

VPC

vpc-0fcb8ea5e50847cb4 | 233523A-vpc

Owner ID

251772956993

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (3)

Edit subnet associations

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
233523A-Subnet-2	subnet-00d94324e495e9c85	10.0.32.0/24	-
233523A-subnet-public1-us-...	subnet-026158b18b453a4c8	10.0.0.0/20	-
233523A-subnet-public2-us-...	subnet-081623510f4e943a1	10.0.16.0/20	-

Subnets without explicit associations (0)

Edit subnet associations

CloudShell

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233523A-load-balancer

aws

Services

Search

[Option+S]

N. Virginia

voclabs/user3238427=MIN_PHYO_THURA @ 2517-7295-6993

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Auto Scaling

Auto Scaling Groups

Settings

EC2 > Load balancers > 233523A-load-balancer

233523A-load-balancer

Details

Load balancer type

Application

Status

Active

VPC

vpc-0fcb8ea5e50847cb4

Load balancer IP address type

IPv4

Scheme

Internet-facing

Hosted zone

Z35SXDOTRQ7X7K

Availability Zones

subnet-00d94324e495e9c85 us-east-1e (use1-az3)

subnet-026158b18b453a4c8 us-east-1a (use1-az6)

subnet-081623510f4e943a1 us-east-1b (use1-az1)

Date created

August 6, 2024, 12:53 (UTC+08:00)

Load balancer ARN

arn:aws:elasticloadbalancing:us-east-1:251772956993:loadbalancer/app/233523A-load-balancer/77f32fbf5a43e34f

DNS name

233523A-load-balancer-1759179830.us-east-1.elb.amazonaws.com (A Record)

CloudShell

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233523A-Auto-Scaling-Group

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

▼ Load Balancing

Load Balancers

Target Groups

Trust Stores [New](#)

▼ Auto Scaling

Auto Scaling Groups

Settings

EC2 > [Auto Scaling groups](#) > 233523A-Auto-Scaling-Group

233523A-Auto-Scaling-Group

Details

Activity

Automatic scaling

Instance management

Monitoring

Instance refresh

Group details

Auto Scaling group name

233523A-Auto-Scaling-Group

Desired capacity

2

Desired capacity type

Units (number of instances)

Amazon Resource Name (ARN)

arn:aws:autoscaling:us-east-1:251772956993:autoScalingGroup:44834e3c-507f-4f64-ad2f-29b1849eb36c:autoScalingGroup/233523A-Auto-Scaling-Group

Date created

Tue Aug 06 2024 13:53:47 GMT+0800 (Singapore Standard Time)

Minimum capacity

2

Status

-

Maximum capacity

6

Launch template

Launch template

lt-0b86b51f67db041dd

233523A-LabConfig-AMI-Template

AMI ID

ami-0d780d7d3c91f4c03

Instance type

t2.micro

Owner

arn:aws:sts::251772956993:assumed-role/voclabs/user3238427=MIN_PHYO_THURA

CloudShell

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Volumes

Snapshots

Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

▼ Load Balancing

Load Balancers

Target Groups

Trust Stores [New](#)

▼ Auto Scaling

Auto Scaling Groups

Settings

Launch template

Launch template

lt-0b86b51f67db041dd

233523A-LabConfig-AMI-Template

AMI ID

ami-0d780d7d3c91f4c03

Instance type

t2.micro

Owner

arn:aws:sts::251772956993:assumed-role/voclabs/user3238427=MIN_PHYO_THURA

Version

Default

Security groups

-

Security group IDs

sg-0e84a4f8e47a92046

Create time

Tue Aug 06 2024 13:49:17 GMT+0800 (Singapore Standard Time)

Description

-

Storage (volumes)

-

Key pair name

233523A-Key_Pair

Request Spot Instances

No

View details in the launch template console

Network

Availability Zones

us-east-1a, us-east-1b, us-east-1e

Subnet ID

subnet-00d94324e495e9c85, subnet-026158b18b453a4c8, subnet-081623510f4e943a1

CloudShell

Feedback

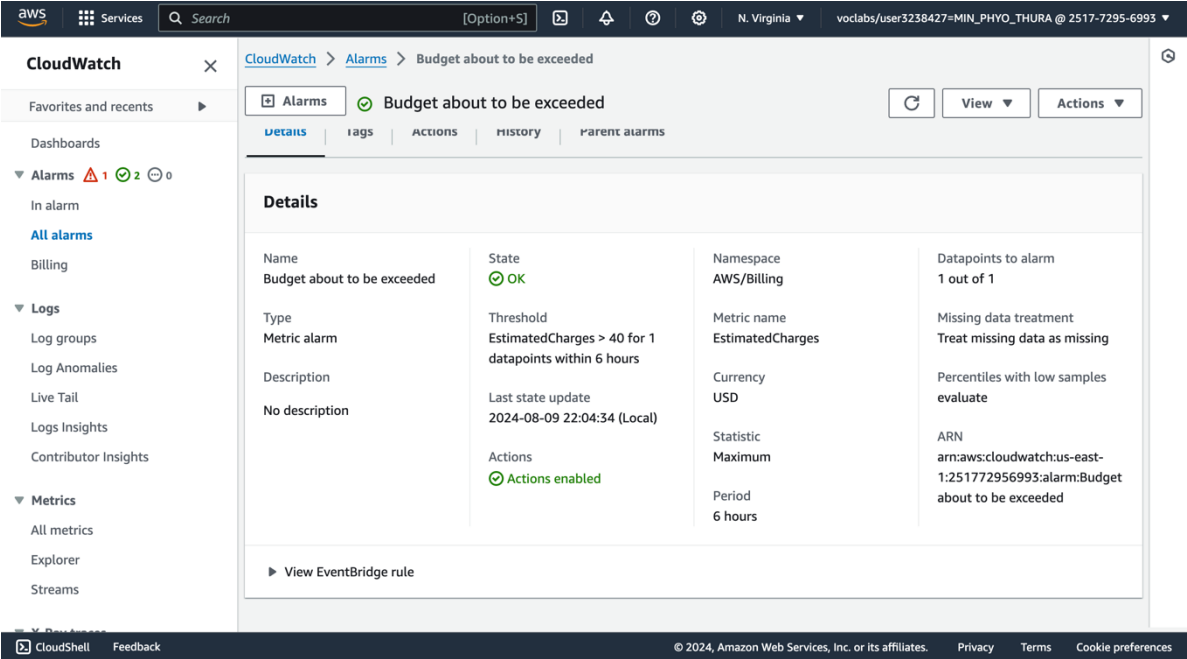
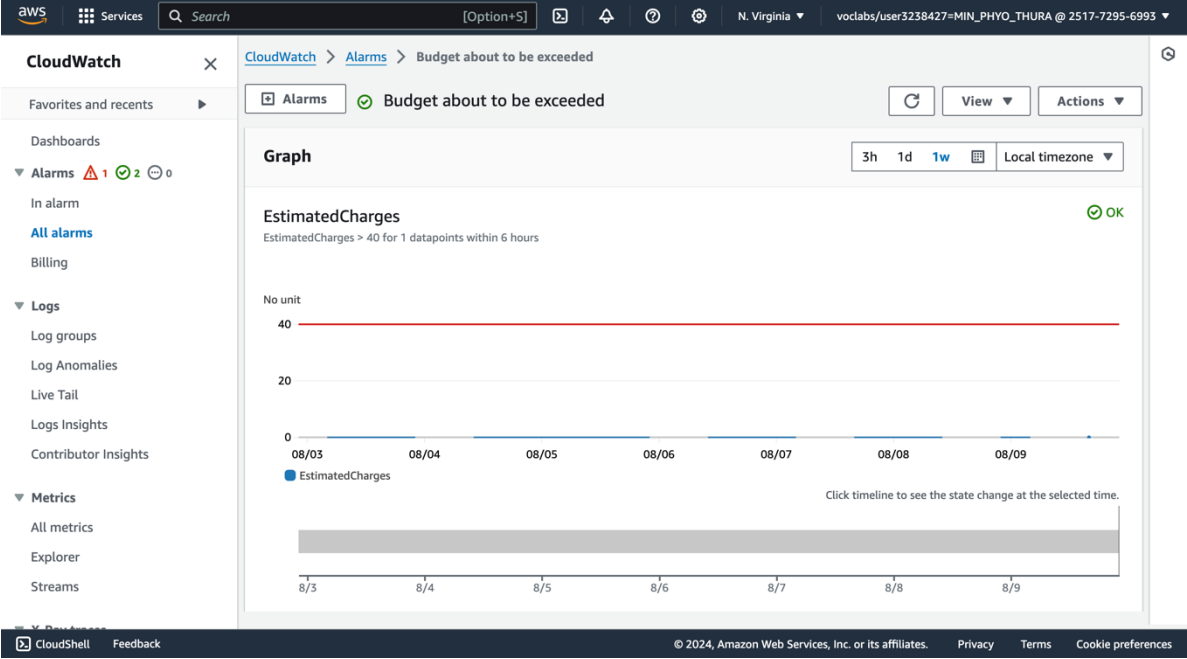
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AWS CloudWatch Alarms



AWS IAM Roles [not allowed]

awsServicesSearch[Option+S]

Failed to create role 233523A-EC2-Role.
User: arn:aws:sts::251772956993:assumed-role/voclabs/user3238427=MIN_PHYO_THURA is not authorized to perform: iam:CreateRole on resource: arn:aws:iam::251772956993:role/233523A-EC2-Role because no identity-based policy allows the iam:CreateRole action

Step 2: Add permissions

Permissions policy summary

Policy name	Type	Attached as
AmazonRDSDataFullAccess	AWS managed	Permissions policy
AmazonS3ReadOnlyAccess	AWS managed	Permissions policy

Step 3: Add tags

Add tags - optional
Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

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AWS CloudFront [not allowed]

awsServicesSearch[Option+S]

Failed to load distributions: User: arn:aws:sts::251772956993:assumed-role/voclabs/user3238427=MIN_PHYO_THURA is not authorized to perform: cloudfront:ListDistributions because no identity-based policy allows the cloudfront:ListDistributions action

CloudFront

Distributions

Policies

Functions

What's new

Telemetry

Monitoring

Alarms

Logs

Reports & analytics

Cache statistics

Popular objects

Top referrers

Usage

Viewers

Security

Origin access

Field-level encryption

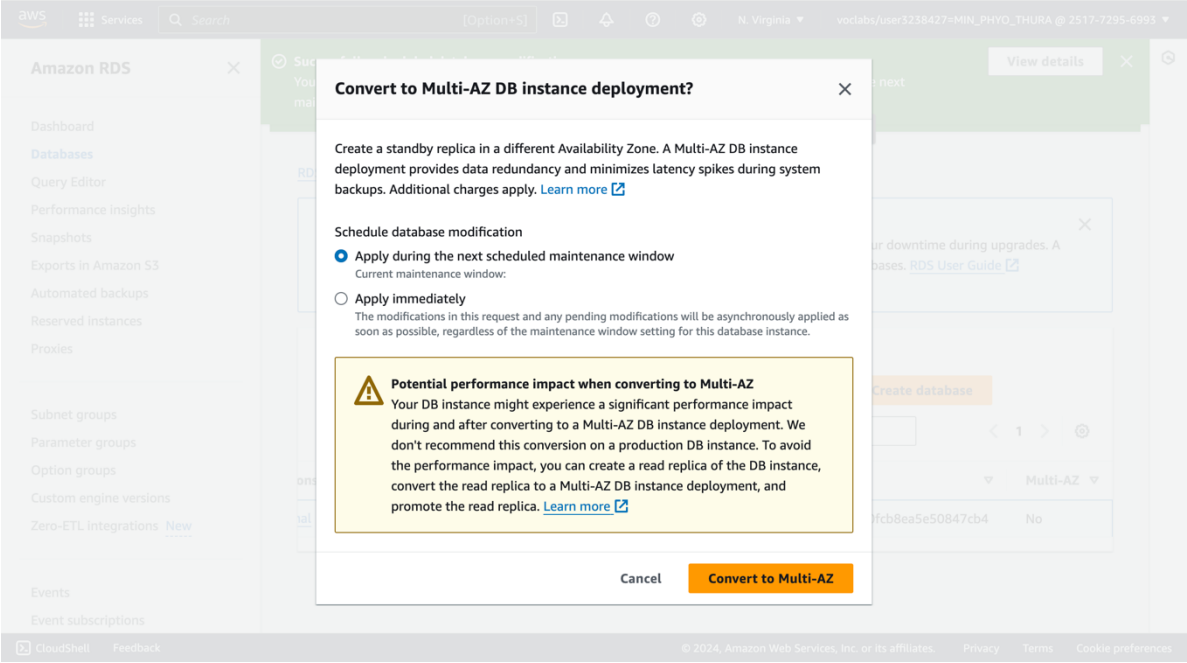
CloudFront > Distributions

Distributions (0)
Search all distributions

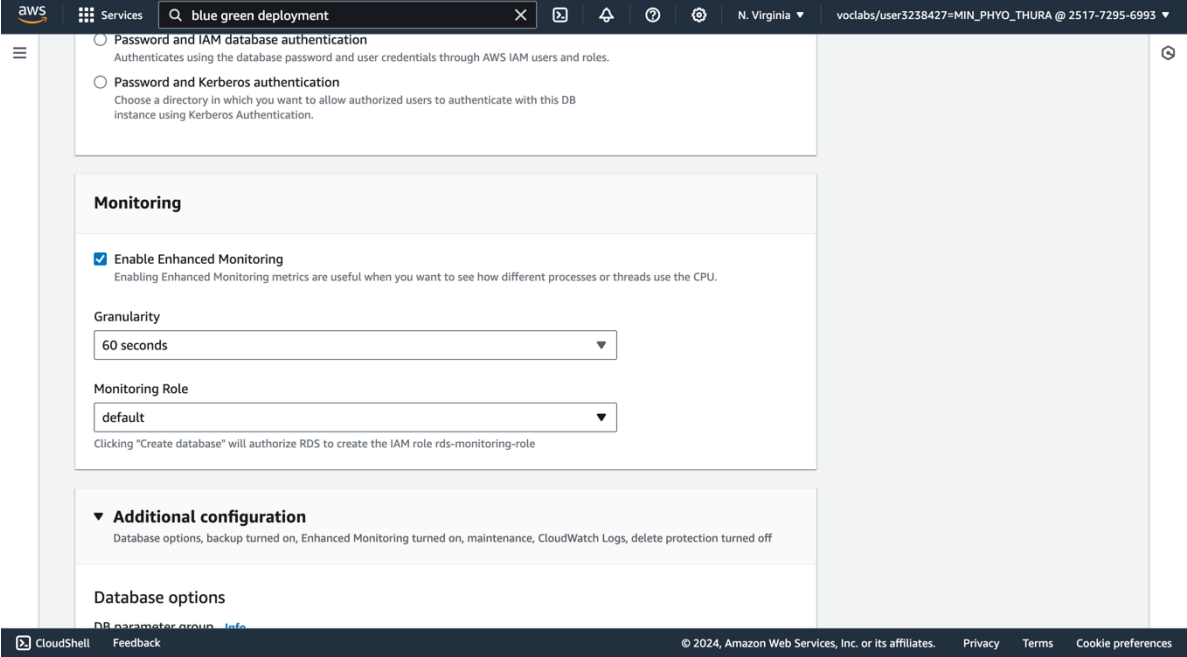
ID	Descript...	Type	Domain ...	Alternat...	Origins
No distributions You don't have any distributions.					

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RDS Multi-AZ instance deployment



Enhanced Monitoring for RDS [not allowed]



Project / EGT208

Admin No. 233523A

aws

Services

blue green deployment

N. Virginia

voclabs/user3238427=MIN_PHYO_THURA @ 2517-7295-6993

RDS service-linked role

Maintenance

Auto minor version upgrade [Info](#)

☒

Enable auto minor version upgrade

Enabling auto minor version upgrade will automatically upgrade to new minor versions as they are released. The automatic upgrades occur during the maintenance window for the database.

DB instance maintenance window

The weekly time range during which system maintenance can occur.

Start day

Monday

Start time

07 : 35 UTC

Duration

0.5 hours

Deletion protection

☐

Enable deletion protection

Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

We're sorry, your request to modify DB instance database-233523a has failed.

User: arn:aws:sts::251772956993:assumed-role/voclabs/user3238427=MIN_PHYO_THURA is not authorized to perform: iam:CreateRole on resource: arn:aws:iam::251772956993:role/rds-monitoring-role because no identity-based policy allows the iam:CreateRole action

Cancel

Continue

CloudShell

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AWS Lambda [not allowed]

aws

Services

Search

[Option+S]

N. Virginia

voclabs/user3238427=MIN_PHYO_THURA @ 2517-7295-6993

Role creation might take a few minutes. Please do not delete the role or edit the trust or permissions policies in this role.

Role name

Enter a name for your new role.

233523A-lambda-role

Use only letters, numbers, hyphens, or underscores with no spaces.

Policy templates - optional [Info](#)

Choose one or more policy templates.

Advanced settings

User: arn:aws:sts::251772956993:assumed-role/voclabs/user3238427=MIN_PHYO_THURA is not authorized to perform: iam:CreateRole on resource: arn:aws:iam::251772956993:role/service-role/233523A-lambda-role because no identity-based policy allows the iam:CreateRole action

Diagnose with Amazon Q

Cancel

Create function

CloudShell

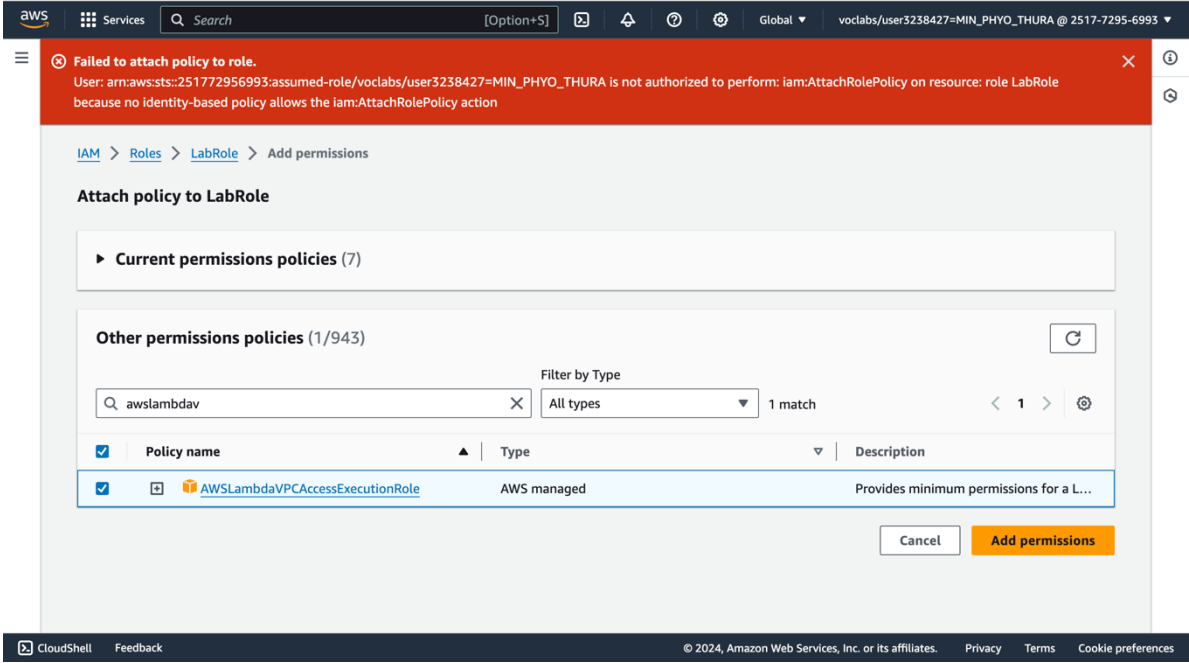
Feedback

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Appendix B

Author/Creator. (Year). Title of the image. Source. URL.

ChatGPT. (2024). An image of 4x4 grid with one electronic appliance in each grid. ChatGPT.